

Semester - III			
Sl. No.	Course Code	Course Title	Page No.
1	MA231TC	Linear Algebra and Probability Theory	1
2	ME232TB	Material Science for Engineers	3
3	CV232TA / BT232TD	Environment and Sustainability / Bioinspired Engineering	5 / 7
4	CS233AI	Data Structures and Applications	9
5	CS234AI	Applied Digital Logic Design and Computer Organization	11
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9	MAT149AT	Bridge Course Mathematics	19
10	SLCSP31A	Domain Engineering Skill Lab	20

Semester - IV			
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2025 Scheme template – UG Programs													
B.E in Electronics and Communication Engineering (ECE)													
III SEMESTER													
Sl	Course Code	Course Title	BoS	Teaching & Learning Scheme				Total Hrs per Sem	Examination				Total Credits
				CI		LI	TW & SL		CIE/SEE Dur.(Hrs)	CIE Marks	SEE Marks	Total Marks	
				L	T	P	S						
1	MA231TC	Linear Algebra and Probability Theory	MA	3 (45)	1 (30)	0 (0)	– (45)	120	2/3	100	100	200	4
2	ME232TB	Material Science for Engineers	ME	3 (45)	0 (0)	1 (30)	– (45)	120	1.5/3	150	150	300	4
3	CV232TA / BT232TD	Environment and Sustainability / Bioinspired Engineering	CV / BT	1 (15)	0 (0)	0 (0)	– (15)	30	1/2	80	100	180	1
4	CS233AI	Data Structures and Applications	CS	3 (45)	0 (0)	1 (30)	– (45)	120	1.5/3	150	150	300	4
5	CS234AI	Applied Digital Logic Design and Computer Organization	CS	3 (45)	0 (0)	1 (30)	– (45)	120	1.5/3	150	150	300	4
6	CS235AT	Operating Systems	CS	3 (45)	0 (0)	0 (0)	– (45)	90	2/3	100	100	200	3
7	CS236AT	Quantum Computing Essentials	CS	3 (45)	0 (0)	0 (0)	– (45)	90	2/3	100	100	200	3
8	CS238DL	Applied Design thinking lab	CS	0 (0)	0 (0)	1 (30)	– (0)	30	2/3	100	100	200	1

III SEMESTER (Continued)													
Sl	Course Code	Course Title	BoS	Teaching & Learning Scheme				Total Hrs per Sem	Examination				Total Credits
				CI		LI	TW & SL		CIE/SEE Dur.(Hrs)	CIE Marks	SEE Marks	Total Marks	
				L	T	P	S						
9	MAT149AT	Bridge Course Mathematics	MA	3 (45)	0 (0)	0 (0)	– (45)	90	2/—	100	—	100	0 [†]
10	SLCSP31A	Domain Engineering Skill Lab	CS	0 (0)	0 (0)	1 (30)	– (0)	30	2/3	100	100	200	1
Total													25

2025 Scheme template – UG Programs													
B.E in Electronics and Communication Engineering (ECE)													
IV SEMESTER													
Sl	Course Code	Course Title	BoS	Teaching & Learning Scheme				Total Hrs per Sem	Examination				Total Credits
				CI		LI	TW & SL		CIE/SEE Dur.(Hrs)	CIE Marks	SEE Marks	Total Marks	
				L	T	P	S						
1	CS241AT	Discrete mathematical structures and combinatorics	CS	3 (45)	0 (0)	0 (0)	– (45)	90	2/3	100	100	200	3
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	CV / ME / BT	3 (45)	0 (0)	0 (0)	– (45)	90	2/3	100	100	200	3

Audit

IV SEMESTER (Continued)													
Sl	Course Code	Course Title	BoS	Teaching & Learning Scheme				Total Hrs per Sem	Examination				Total Credits
				CI		LI	TW & SL		CIE/SEE Dur. (Hrs)	CIE Marks	SEE Marks	Total Marks	
				L	T	P	S						
3	CD343AI	Design and Analysis of Algorithms	CD	3 (45)	0 (0)	1 (30)	– (45)	120	1.5/3	150	150	300	4
4	CS345AI	Embedded Systems, IoT and Edge AI	CS	3 (45)	0 (0)	1 (30)	– (45)	120	1.5/3	150	150	300	4
5	CS246AT	Computer Networks	CS	3 (45)	1 (30)	0 (0)	– (45)	120	2/3	100	100	200	4
6	CS248ATX	Professional Electives	CS	3 (45)	0 (0)	0 (0)	– (45)	90	2/3	100	100	200	3
7	HS248LX	AE Course	HS	0 (0)	0 (0)	1 (30)	– (0)	30	2/3	100	100	200	1
8	CS139AT	Bridge Course C Programming	CS	3 (45)	0 (0)	0 (0)	– (45)	90	2/—	100	—	100	0 [†]
Total													22

Audit

Semester - IV			
Professional Electives			
GROUP-A			
Sl. No.	Course Code	Course Title	Page No.
1	CS248ATA	Graph Theory	30
2	CS248ATB	Computer Graphics and Extended Reality	32
3	CI248ATC	Quantum Algorithms and Key Distribution	34
4	CS248ATD	Natural Language Processing	36
5	CS355TBF	Linux Internals	38
6	CS365TDB	Operation Research	40

Semester - IV			
AE Courses			
Sl. No.	Course Code	Course Title	Page No.
1	HS238LA	National Service Scheme	42
2	HS238LB	National Cadet Corps	44



Semester - III					
Linear Algebra and Probability Theory Category: Professional Core Elective (Common to Common to CS and CY) (Theory & Tutorial)					
Course Code	:	MA231TC		CIE	: 100 Marks
Credit distribution: L:T:P	:	3:1:0		SEE	: 100 Marks
Pedagogical Hours (L:T:P:S)	:	45:30:0:45		CIE Duration	: 2 Hours
Total credits:	:	4		SEE Duration	: 3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - III					
Material Science for Engineers					
Category: Professional Core Course					
(Common to CS,CD,CY and CI Programs)					
(Theory & Practical)					
Course Code	:	ME232TB		CIE	: 100 + 50 Marks
Credit distribution: L:T:P	:	3:0:1		SEE	: 100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45		CIE Duration	: 1.5 Hours
Total credits:	:	4		SEE Duration	: 3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40



RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (Continued)		
#	COMPONENTS	MARKS
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - III					
Environment and Sustainability					
Category: Professional Core Course					
(Common to CS,CD,CY and CI Programs)					
(Theory)					
Course Code	:	CV232TA		CIE	: 80 Marks
Credit distribution: L:T:P	:	1:0:0		SEE	: 100 Marks
Pedagogical Hours (L:T:P:S)	:	15:0:0:15		CIE Duration	: 1 Hours
Total credits:	:	1		SEE Duration	: 2 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Compulsory	18
5 & 6	Unit - III: Compulsory	18
7 & 9	Unit - IV: Compulsory	18
9 & 10	Unit - V: Compulsory	18
TOTAL		100



Semester - III					
Bioinspired Engineering					
Category: Professional Core Course					
(Common to CS,CD,CY and CI Programs)					
(Theory)					
Course Code	:	BT232TD		CIE	: 80 Marks
Credit distribution: L:T:P	:	1:0:0		SEE	: 100 Marks
Pedagogical Hours (L:T:P:S)	:	15:0:0:15		CIE Duration	: 1 Hours
Total credits:	:	1		SEE Duration	: 2 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Compulsory	18
5 & 6	Unit - III: Compulsory	18
7 & 9	Unit - IV: Compulsory	18
9 & 10	Unit - V: Compulsory	18
TOTAL		100



Semester - III						
Data Structures and Applications Category: Professional Core Course (Common to Common to CS and CY) (Theory & Practical)						
Course Code	:	CS233AI		CIE	:	100 + 50 Marks
Credit distribution: L:T:P	:	3:0:1		SEE	:	100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45		CIE Duration	:	1.5 Hours
Total credits:	:	4		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design						
Innovative Experiments 2. [EL] For EL experiment						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40



RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (Continued)		
#	COMPONENTS	MARKS
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - III						
Applied Digital Logic Design and Computer Organization						
Category: Professional Core Course						
(Common to Common to CS, CD and CY)						
(Theory & Practical)						
Course Code	:	CS234AI		CIE	:	100 + 50 Marks
Credit distribution: L:T:P	:	3:0:1		SEE	:	100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45		CIE Duration	:	1.5 Hours
Total credits:	:	4		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Hardware experiments						
Here you can put your instructions here. If you don't want this instruction, simply comment this line.						
1. Design						
Innovative Experiments						
2. [EL] For EL experiment						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40



RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (Continued)		
#	COMPONENTS	MARKS
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - III						
Operating Systems						
Category: Professional Core Course						
(Common to Common to CS, CD and CI)						
(Theory)						
Course Code	:	CS235AT		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - III						
Quantum Computing Essentials						
Category: Professional Core Course						
(Common to Common to CS, CD and CI)						
(Theory)						
Course Code	:	CS236AT		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - III					
Applied Design thinking lab Category: Professional Core Course (Common to Common to CS,CD,CY and CI Programs) (Practical)					
Course Code	:	CS238DL		CIE	: 50 + 50 Marks
Credit distribution: L:T:P	:	0:0:1		SEE	: 50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0		CIE Duration	: 2 Hours
Total credits:	:	1		SEE Duration	: 3 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80



RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Semester - III						
Bridge Course Mathematics						
Category: Bridge Course						
(Common to Common to CS,CD,CY and CI Programs)						
(Theory)						
Course Code	:	MAT149AT		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



Semester - III					
Domain Engineering Skill Lab					
Category: Skill Lab					
(Common to Common to CS,CD,CY and CI Programs)					
(Practical)					
Course Code	:	SLCSP31A		CIE	: 50 + 50 Marks
Credit distribution: L:T:P	:	0:0:1		SEE	: 50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0		CIE Duration	: 2 Hours
Total credits:	:	1		SEE Duration	: 3 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80



RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Semester - IV						
Discrete mathematical structures and combinatorics						
Category: Professional Core Course						
(Common to Common to CS and CY Programs)						
(Theory)						
Course Code	:	CS241AT		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
Design and Analysis of Algorithms Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Theory & Practical)					
Course Code	:	CD343AI		CIE	: 100 + 50 Marks
Credit distribution: L:T:P	:	3:0:1		SEE	: 100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45		CIE Duration	: 1.5 Hours
Total credits:	:	4		SEE Duration	: 3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40



RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (Continued)		
#	COMPONENTS	MARKS
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - IV					
Embedded Systems, IoT and Edge AI Category: Professional Core Course (Common to CS and CI Programs) (Theory & Practical)					
Course Code	:	CS345AI		CIE	: 100 + 50 Marks
Credit distribution: L:T:P	:	3:0:1		SEE	: 100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45		CIE Duration	: 1.5 Hours
Total credits:	:	4		SEE Duration	: 3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to: 1. sample					
References [1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40



RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (Continued)		
#	COMPONENTS	MARKS
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - IV					
Computer Networks					
Category: Professional Core Course					
(Common to CS,CD,CY and CI Programs)					
(Theory & Tutorial)					
Course Code	:	CS246AT		CIE	: 100 Marks
Credit distribution: L:T:P	:	3:1:0		SEE	: 100 Marks
Pedagogical Hours (L:T:P:S)	:	45:30:0:45		CIE Duration	: 2 Hours
Total credits:	:	4		SEE Duration	: 3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV						
Graph Theory						
Category: Professional Core Elective						
(Only CS)						
(Theory)						
Course Code	:	CS248ATA		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV						
Computer Graphics and Extended Reality						
Category: Professional Core Elective						
(Only CS)						
(Theory)						
Course Code	:	CS248ATB		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV						
Quantum Algorithms and Key Distribution						
Category: Professional Core Elective						
(Common to CS, CI and CD Programs)						
(Theory)						
Course Code	:	CI248ATC		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV						
Natural Language Processing						
Category: Professional Core Elective						
(Only CS)						
(Theory)						
Course Code	:	CS248ATD		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV						
Linux Internals						
Category: Professional Core Elective						
(Common to CS, CI, CY and CD Programs)						
(Theory)						
Course Code	:	CS355TBF		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV						
Operation Research						
Category: Professional Core Elective						
(Only CS)						
(Theory)						
Course Code	:	CS365TDB		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
National Service Scheme					
Ability Enhancement Course					
(All programs)					
(Practical)					
Course Code	:	HS238LA		CIE	: 50 + 50 Marks
Credit distribution: L:T:P	:	0:0:1		SEE	: 50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0		CIE Duration	: 2 Hours
Total credits:	:	1		SEE Duration	: 3 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80



RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Semester - IV					
National Cadet Corps					
Ability Enhancement Course					
(All programs)					
(Practical)					
Course Code	:	HS238LB		CIE	: 50 + 50 Marks
Credit distribution: L:T:P	:	0:0:1		SEE	: 50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0		CIE Duration	: 2 Hours
Total credits:	:	1		SEE Duration	: 3 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80



RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Semester - IV						
Bridge Course C Programming						
Category: Bridge Course						
(Common to Common to CS,CD,CY and CI Programs)						
(Theory)						
Course Code	:	CS139AT		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	3 Hours
Unit - I						4 Hours
sample						
Course Outcomes: After completing the course, the students will be able to:						
1. sample						
References						
[1] Razavi2000						

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100